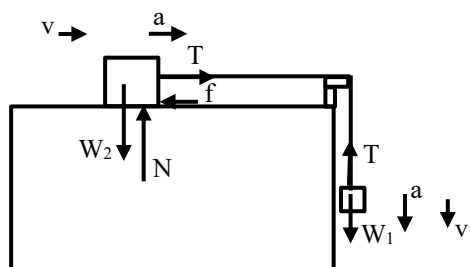


## Group Problem 2 Solution

Useful Sketch



Know:

$W_1$   
 $W_2$   
 $f$

$a$ : acceleration of each block  
 $T$ : pull of the rope on each block  
 $W_1$ : weight of block 1 (hangs vertically)  
 $W_2$ : weight of block 2 (on ramp)  
 $N$ : normal force of the surface on block 2

Question: What is the acceleration of block 2?

Physics approach:

Since this problem involves forces and their relation to acceleration use  $\sum \vec{F} = m\vec{a}$

Add forces independently on each block.

The magnitude of the acceleration of each block is the same because they are attached together.

The magnitude of the velocity of each block is the same because they are attached together.

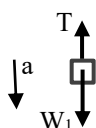
Assume that the rope's weight can be neglected relative to either block's weight.

Assume that there is no friction on the rope going over the rail because it is slippery.

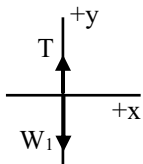
The same rope connects both blocks. The force the rope exerts on block 2 is caused by the force block 1 exerts on the rope.

By Newton's 3<sup>rd</sup> law, the force of block 1 on the rope is equal to the force of the rope on block 1. The force the rope exerts on each block has the same magnitude, the rope tension.

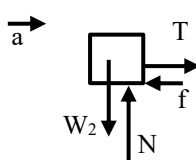
Free Body Diagram  
Block 1



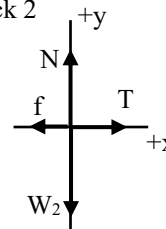
Force Diagram  
Block 1



Free Body Diagram  
Block 2



Force Body Diagram  
Block 2



Know:  $f$ ,  $W_1 = m_1g$ ,  $W_2 = m_2g$

Block 1 (from force diagram)

$$\sum F_x = 0$$

$$\sum F_y = T - W_1$$

$$m_1(-a) = T - W_1$$

$$W_1 = m_1g$$

Block 2 (from force diagram)

$$\sum F_x = T - f$$

$$m_2a = T - f$$

$$\sum F_y = N - W_2 = 0$$

$$0 = N - W_2$$

$$W_2 = m_2g$$

Plan:

Determine the rope force,  $T$ , from the behavior of block 1.

Since the same rope connects block 2, use the same rope force to determine the acceleration of block 2.

Target: a as a function of f

PLAN the Math

|                                            |     | unknowns |
|--------------------------------------------|-----|----------|
| Find a                                     |     | a        |
| $m_2 a = T - f$                            | [1] | T, $m_2$ |
| Find T                                     |     |          |
| $m_1(-a) = T - W_1$                        | [2] | $m_1$    |
| Find $m_1$                                 |     |          |
| $W_1 = m_1 g$                              | [3] |          |
| Find $m_2$                                 |     |          |
| $W_2 = m_2 g$                              | [4] |          |
| 4 unknowns and 4 equations,<br>OK to solve |     |          |

Do the Math

Solve [4] for  $m_2$

$$m_2 = \frac{W_2}{g} \quad \text{put into 1}$$

$$\frac{W_2}{g} a = T - f \quad [1]$$

Solve [3] for  $m_1$

$$m_1 = \frac{W_1}{g} \quad \text{put into 2}$$

$$-\frac{W_1}{g} a = T - W_1 \quad [2]$$

Solve [2] for T

$$W_1 - \frac{W_1}{g} a = T \quad \text{put into 1}$$

$$\frac{W_2}{g} a = W_1 - \frac{W_1}{g} a - f$$

Solve for a

Get all terms with a on one side of the equation.

$$\frac{W_2}{g} a + \frac{W_1}{g} a = W_1 - f$$

Factor out a

$$a \left( \frac{W_2}{g} + \frac{W_1}{g} \right) = W_1 - f$$

Divide both sides of the equation by  $\left( \frac{W_2}{g} + \frac{W_1}{g} \right)$

$$a = \frac{W_1 - f}{\left( \frac{W_2}{g} + \frac{W_1}{g} \right)}$$

Simplifying

$$a = \frac{W_1 - f}{\frac{1}{g}(W_2 + W_1)}$$

$$a = \frac{W_1 - f}{(W_2 + W_1)} g$$

EVALUATE the ANSWER

Check units:

$g$  has units of acceleration and  $W_1$ ,  $W_2$ , and  $f$  have units force.

$$\frac{[\text{force}] - [\text{force}]}{([\text{force}] + [\text{force}])} [\text{acceleration}] = [a]$$

$$\frac{[\text{force}]}{[\text{force}]} [\text{acceleration}] = [a]$$

The force units cancel. The units of  $a$  are acceleration which is correct. The units check.

Is Result Unreasonable?

As the frictional force increases, the numerator gets smaller so the acceleration of the heavy block decreases. This is reasonable since higher friction would make it more difficult to change the motion of the block.

If the weight of the hanging block were equal to the frictional force, the block would move at a constant velocity. This would make it easier to control for construction purposes so that behavior is reasonable.

If the weight of the heavy block increases, the acceleration decreases. This is reasonable since it is more difficult to change the motion of a heavier object.